

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

End Semester Examination – Summer23

Date: 19/07/2023

Course: B. Pharmacy

Sem: VI

Subject Name: Pharmaceutical Biotechnology

Subject Code:BP605T

Max Marks: 75

Duration:

3 Hr.

Instructions:

- 1. All questions are compulsory**
 - 2. Draw diagrams / figures wherever necessary**
 - 3. Figures to right indicate full marks**
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Q. 1. Objective Type Questions (Answer all the questions) (10 x 2) = 20

- i) What are plasmids? Give one example.
- ii) What is protein engineering?
- iii) Write the classification of hypersensitivity reactions.
- iv) Define terms: a) Transcription b) Translation
- v) Define antigen and antibody..
- vi) Differentiate the Vaccine and toxoid
- vii) Define vector. Write the features of a good vector.
- viii) What is gene therapy?
- ix) Enlist the enzymes used in DNA cloning
- x) Name four mutagenic agents.

Q. 2. Long Answers (Answer 2 out of 3) (2 x 10) = 20

- i) Explain in details about production and purification of monoclonal antibodies by hybridoma technology
- ii) Describe the general method of recombinant DNA technology.
- iii) Explain different methods of enzyme immobilization with their advantages and disadvantages

Q. 3. Short Answers (Answer 7 out of 9) (7 x 5) = 35

- i) What are biosensors? Write its principle and its functions.
- ii) Write a detailed note on cloning vectors.
- iii) What is PCR? Write the working principle of PCR.
- iv) Write in brief the Collection, Processing and Storage of whole human blood.
- v) Write in brief the southern blot technique and mention its applications.
- vi) Explain different types of mutations.
- vii) Write a note on microbial biotransformation in production of steroidal medicinal agents.
- viii) Describe the structure and functions of MHC.
- ix) Explain the production of Penicillin by fermentation technology with a neat labelled flow chart.

-----END OF THE PAPER-----